

Professional Profile

Nirmal Nipuna Nanayakkara

Senior Front-End Engineer | UI/UX Specialist | Angular Expert | AI Application Developer

I am a highly accomplished and passionate Senior Front-End Engineer and UI/UX Specialist with more than 12 years of professional experience designing, architecting, and delivering scalable enterprise-grade web and mobile applications. Throughout my career, I have consistently demonstrated the ability to transform complex business requirements into intuitive, user-centric digital experiences that drive measurable business outcomes.

My expertise spans the complete software development lifecycle, from product ideation and user research through architecture design, development, deployment, optimization, and maintenance. I have successfully contributed to and led the development of numerous enterprise solutions across industries including Logistics, Healthcare, Beauty, Food Distribution, E-Learning, Artificial Intelligence, Cognitive Computing, and Community Platforms.

Over the years, I have developed deep expertise in Front-End Engineering, User Interface Design, User Experience Design, Single Page Applications (SPAs), Micro-Frontend Architectures, Mobile Application Development, and AI-powered applications. My ability to combine technical excellence with design thinking allows me to create products that are not only visually appealing but also highly functional, scalable, and user-friendly.

Technical Leadership and Front-End Expertise

My primary specialization is modern front-end development using Angular, where I have extensive experience building complex enterprise applications from scratch. I have led front-end architecture decisions, established coding standards, designed reusable component libraries, and implemented scalable development practices that improve maintainability and team productivity. My experience includes designing and developing:

- Enterprise-grade web applications
- SaaS platforms
- AI-powered user interfaces
- Progressive Web Applications (PWAs)
- Mobile applications
- Single Page Applications (SPAs)
- Micro-Frontend ecosystems
- Component-based design systems
- Internal business platforms
- Data visualization dashboards
- Real-time applications

I am particularly skilled at creating maintainable front-end architectures that enable teams to scale applications efficiently while maintaining high performance and code quality.

User Experience and Product Design Excellence

My background in UI/UX development has provided me with a unique perspective that bridges the gap between design and engineering. I understand that successful products are not built solely through technical implementation but through a deep understanding of user behavior, business objectives, and design principles.

I have extensive experience in:

- User Interface Design
- User Experience Design
- Wireframing
- Interactive Prototyping
- Design Systems
- Information Architecture
- Responsive Design
- Mobile-First Design
- Accessibility Standards (WCAG)
- Design Thinking

- User Journey Mapping
- Interaction Design

Throughout my career, I have translated business requirements into wireframes, mockups, and production-ready interfaces that enhance user satisfaction while supporting business growth objectives.

Artificial Intelligence and Emerging Technologies

I am passionate about emerging technologies and have actively contributed to AI-driven projects and intelligent user experiences. My experience includes developing interfaces for AI applications, conducting research into next-generation human-computer interactions, and integrating AI-powered features into enterprise solutions.

At the Colombo Centre for Cognitive Computing, I worked on advanced AI interfaces and humanoid robot interaction systems, focusing on creating natural and engaging user experiences that improve communication between humans and intelligent systems.

I actively leverage modern AI tools including:

- Claude AI
- OpenAI
- Codex
- Microsoft AI Foundry

These technologies enable me to accelerate development workflows, generate technical documentation, improve code quality, assist with debugging, and create innovative user experiences.

Software Engineering Practices

I strongly believe that successful software development requires more than coding expertise. Throughout my career, I have consistently promoted engineering excellence through:

- Clean Code Principles
- SOLID Design Patterns
- Component-Based Architecture
- Code Reviews
- Technical Documentation
- Test-Driven Development
- Continuous Integration and Deployment
- Agile Methodologies
- Scrum Practices
- Performance Optimization
- Accessibility Compliance
- Security Best Practices

I have extensive experience collaborating with cross-functional teams consisting of product managers, UX designers, QA engineers, backend developers, DevOps engineers, and business stakeholders to ensure successful project delivery.

Cloud, DevOps, and Development Tools

I am proficient with modern development ecosystems and cloud technologies including:

Cloud Platforms

- Microsoft Azure
- Amazon Web Services (AWS)
- Firebase

Development Tools

- Git
- Bitbucket
- GitHub

- NPM
- Webpack
- Grunt
- Gulp

Project Management

- JIRA
- Agile Scrum Frameworks

Design Tools

- Figma
- Adobe XD

Backend & Integration Technologies

- Node.js
- Python
- .NET
- GraphQL
- REST APIs
- WebSockets
- MySQL
- Firebase Services

Leadership and Team Development

Beyond technical expertise, I am passionate about helping teams grow and succeed. Throughout my career, I have mentored junior developers, conducted knowledge-sharing sessions, performed code reviews, and established development standards that improve both individual and team performance.

I enjoy:

- Mentoring developers
- Conducting technical workshops
- Leading architecture discussions
- Reviewing code and enforcing best practices
- Sharing industry knowledge
- Supporting career development initiatives
- Building collaborative engineering cultures

I believe that knowledge sharing is one of the most important responsibilities of experienced engineers. Much of my own growth has been made possible through learning from talented professionals, and I strive to provide the same opportunities for others.

Professional Strengths

My key strengths include:

- Front-End Architecture Design
- Angular Enterprise Development
- UI/UX Design and Implementation
- SaaS Product Development
- AI Application Development
- Problem Solving and Critical Thinking
- Performance Optimization
- Technical Leadership
- Team Collaboration
- Agile Delivery
- Client Communication
- Stakeholder Management
- Mentorship and Coaching
- Attention to Detail

- Time Management
- Continuous Learning

Career Objective

I am seeking opportunities to contribute my expertise in front-end engineering, UI/UX design, SaaS product development, and AI-driven applications within innovative organizations that value technical excellence, creativity, and continuous improvement.

My goal is to leverage my extensive experience to build impactful digital products, lead high-performing engineering initiatives, and help organizations create exceptional user experiences that drive measurable business success while continuing to grow as a technology leader.

Education & Professional Certifications

Professional Certifications

Certified Software Engineer

HackerRank - hackerrank.com/certificates/be02598e6818

Successfully earned the Software Engineer Certification by demonstrating proficiency in:

- Software Development Best Practices
- Problem Solving
- Algorithms and Data Structures
- Software Architecture
- Clean Coding Principles
- Software Engineering Fundamentals

This certification validates advanced software engineering capabilities and practical coding expertise required to solve real-world engineering challenges.

Angular Certification

Udemy - udemy.com/certificate/UC-50121f4e-4677-4003-a0eb-93518043fbae/

Completed advanced Angular training focused on enterprise-level application development, including:

- Angular Architecture
- Components and Modules
- Dependency Injection
- Routing and Navigation
- Reactive Forms
- RxJS
- State Management
- REST API Integration
- Performance Optimization
- Enterprise Application Development

This certification further strengthened expertise in building scalable, maintainable, and high-performance Angular applications.

Angular Certification (Hard)

TestDome - testdome.com/certificates/aff0860ee6eb45d7a84c88472d42a421

Successfully passed the advanced Angular assessment designed to evaluate real-world Angular development skills.

Key areas assessed included:

- Angular Framework Architecture
- TypeScript Development
- Reactive Programming
- Component Design
- Dependency Injection
- Application Optimization
- Performance Best Practices
- Enterprise Front-End Development

This certification demonstrates strong technical competency and practical expertise in modern Angular application development.

Continuous Professional Development

As a technology professional, I strongly believe in continuous learning and staying ahead of industry trends. I actively invest time in expanding my knowledge across:

- Artificial Intelligence
- Generative AI Technologies
- Front-End Architecture
- Cloud Computing
- Modern JavaScript Ecosystem
- UX Research & Design Systems
- Performance Engineering
- Accessibility Standards (WCAG)
- Mobile Application Development
- Enterprise Software Architecture

I regularly leverage industry-leading platforms, technical communities, conferences, documentation, and AI-assisted development tools to remain at the forefront of emerging technologies and software engineering best practices.

Professional Philosophy

Education extends beyond formal qualifications and certifications. Throughout my career, I have maintained a strong commitment to self-learning, experimentation, mentorship, and knowledge sharing. This growth mindset has enabled me to evolve from a UI/UX Developer into a Senior Front-End Engineer capable of leading enterprise-scale projects, mentoring teams, architecting complex solutions, and contributing to AI-powered products used across global organizations.